DS5006 Machine Learning — Project Proposal

IIT Palakkad

Data Pertubations - How it affects Satisfaction Prediction

A1. Team.

Team name: Team Arch

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A2. Title.

Data Pertubations - How it affects Satisfaction Prediction

A3. Data.

- **Source:** Kaggle — *Binary Aviation Satisfaction 129K* (`https://www.kaggle.com/datasets/arseniyshutko/binary-aviation-satisfaction-129k`).
- **Dataset size:** 129,880 passenger records and 22 columns, including the target variable.
- **Input variables:** Passenger characteristics, flight information, service-rating variables, and delay-related variables.
- **Target variable:** Binary passenger *satisfaction*.
- **Data type:** Tabular data containing numerical, categorical, and ordinal variables.

A4. Scientific Questions.

Q1 — Does the numerical meaning of an ordinal rating scale matter?

Hypothesis and rationale. Predictions will not be completely invariant to numerical spacing, as the effect may differ across model classes. Tree-based models primarily use ordering and split thresholds, whereas linear and kernel methods use numerical distances.

Manipulation. For every ordinal service-rating feature, category ordering will be preserved while the original numerical encoding is replaced by strictly increasing alternatives, for example

$$1, 2, 3, 4, 5 \longrightarrow 1, 2, 4, 7, 11.$$

Several pre-specified monotone encodings with increasingly unequal spacing will be used. Raw observations, labels, train/test splits, preprocessing, and hyperparameter selection will otherwise remain fixed. Thus, only the numerical representation is changed.

Measurement. Across repeated stratified train/validation/test splits, balanced accuracy, ROC-AUC, and the paired change in predicted probability for the same passengers will be measured. The fraction of test predictions whose class changes will also be recorded. Results will be reported as means with spread across splits.

No result. The hypothesis is unsupported if transformed encodings produce negligible paired changes in performance and predictions within a pre-specified practical tolerance.

Q2 — Are learned satisfaction predictions monotone under controlled worsening of flight conditions?

Hypothesis and rationale. Many models are expected to assign lower satisfaction probabilities to increasingly severe delays. However, non-monotone reversals may occur for some passengers because the models learn correlations from observational data rather than an explicitly constrained monotone function.

Manipulation. For selected test passengers, all observed characteristics except delay will be held fixed. Controlled counterfactual versions will be created with increasing delay severity, for example

$$d \in \{5, 20, 40, 60, 80, 100\} \text{ minutes.}$$

Where both departure and arrival delay are used, their perturbation will follow a pre-specified consistent rule. No retraining will be performed for these counterfactual inputs. The experiment tests the learned prediction function, not a causal effect.

Measurement. For each passenger and each pair of adjacent delay levels, the predicted satisfaction probabilities will be computed. The primary outcome will be the monotonicity-violation rate.

No result. The hypothesis is unsupported if violation rates remain negligible and similar across delay levels and model classes, indicating effectively monotone learned functions under the tested perturbations.

Q3 — Does the effect of training-data corruption depend on the corruption type and location?

Hypothesis and rationale. Different types of training-data corruption are expected to produce different levels of degradation even when the same fraction of observations is corrupted. Label corruption is expected to be more damaging than mild feature corruption, while corruption concentrated near the decision boundary may disproportionately affect the learned boundary.

Manipulation. Only the training data will be corrupted; validation and test data will remain unchanged. Three corruption mechanisms will be compared:

1. **Weak feature corruption:** replace one selected service-rating value with another valid rating.
2. **Strong feature corruption:** corrupt several or all predictor fields for selected training observations.

Each corruption mechanism will be applied either uniformly or preferentially near a reference decision boundary. The same original training splits and corruption masks will be reused across models to enable paired comparisons.

Measurement. Balanced accuracy and ROC-AUC will be measured for each corruption condition. The primary outcome will be the paired performance drop relative to clean training data at the same corruption level. Means and spread across repeated data splits will be reported.

No result. The hypothesis is unsupported if different corruption types and locations produce practically indistinguishable performance changes at matched corruption levels.

A5. Models.

The central models will be logistic regression, support vector machines (SVM), decision trees and ensemble learning methods. A majority-class classifier will be included as the required simple reference model, and a regularised logistic model will provide a linear reference.

Experimental Protocol.

The same repeated stratified splits will be used for all conditions within each repetition. At least five repetitions will be performed, varying the data split rather than only the model random seed. Hyperparameters will be selected using training/validation data only; each test portion will be evaluated once.

For manipulated conditions, means and spread will be reported and comparisons will use paired differences on identical splits. Random seeds and preprocessing choices will be fixed and recorded.

A6. Risks and Fallbacks.

1. **Q1:** Q1 may produce very small effects because some models depend mainly on ordering. This constitutes a meaningful negative result.
2. **Q2:** Unrealistic counterfactuals - some delay combinations may be unrealistic. Perturbations will therefore be restricted to plausible ranges, and the experiment will be interpreted as a controlled input perturbation rather than a causal claim.
3. **Excessive corruption:** Strong corruption may make the task unrealistically difficult.
4. **Computational cost:** Six model families across multiple repetitions and experimental conditions may be computationally substantial. A fixed experimental grid and common preprocessing pipeline will be used.

A7. Planned Division of Work.

- **Member 1:** Data auditing, preprocessing, Q1 experiment, and analysis.
- **Member 2:** Q2 counterfactual construction, monotonicity metrics, and analysis.
- **Member 3:** Q3 corruption experiments, evaluation, and robustness analysis.